Vocab Bash Time :D [53.5 pts; 0.5 pts each; +3 for each fully correct set (A, B, C, etc.)]

Match each word with the most fitting definition

A. Parts of a Riverine system

a	Lateral Bar	1	Deposited unit found on the inside curve of meanders
b	Anabranch	2	Erosion of riverbed by fast flowing water
c	Point bar	3	Temporary river channel of ephemeral river in xeric environments
d	Fill Terrace	4	Tranquil or standing flow with intervening regions of imbricated bed material near inflection points in meanders
e	Anastomosing river	5	Bank-attached unit predominantly of gravel, found along low-sinuosity reaches; develops along alternate sides of stream
f	Oxbow Lake	6	Lake formed from cutoff meanders
g	Cut Terrace	7	Stepped, flat, elongate bank-attached unit; straight to gently curved planform with same material as floodplain
h	Playa	8	River section that diverts and rejoins
i	Riffle-Pool sequence	9	Flowing water body that diverges and rejoins trunk stream after a significant distance
j	Chute Cutoff	10	Ephemeral lakes found in xeric environments
k	Arroyo	11	Short circuiting a meander, may lead to creation of oxbow lake
l	Perennial stream	12	raised, flat area of depositional origin in a river valley
m	Braided stream	13	raised, flat area of erosional origin in a river valley
n	Ledge	14	Water body that flows year round
o	Yazoo tributary	15	River with high sediment loads and interconnected channels separated by bars
p	Scour	16	River with interconnected channels separated by distinct floodplains

A.

a. 5
b. 8
c. 1
d. 12
e. 16
f. 6
g. 13
h. 10
i. 4
j. 11
k. 3
l. 14
m. 15
n. 7
o. 9
p. 2

B. River Processes

a	Stream Piracy	1	Sediment and pore spaces beneath and along stream bed
b	River Rejuvenation	2	Mechanical erosion via transported material
c	First Flush	3	Water diversion in which a stream is captured by another stream
d	Hyporheic zone	4	Initial surface runoff during precipitation
e	Imbrication	5	Sedimentary material deposited during floods
f	Avulsion	6	Diversion of main path of river
g	Lag deposit	7	Overflow of river from natural channel
h	Flooding	8	Deposit of material winnowed by physical action
i	Crevasse Splay	9	Exhibited by runoff across a large flat region contrasted with gully formation
j	Sheetflow	10	Directional selection of a preferred orientation found in sediments
k	Slackwater deposits	11	Decrease in river base level
l	Corrasion	12	Sedimentary fluvial deposit when a river breaks out of its levees

B.

a. 3
b. 11
c. 4
d. 1
e. 10
f. 6
g. 8
h. 7
i. 12
j. 9
k. 5
l. 2

C. Standing water

a	Euxinia	1	Oxygen poor
b	Bjerrum Plot	2	Lower water layer in a stratified lake
c	Hypereutrophic	3	Water source from precipitation
d	Hypolimnion	4	Well lit area near shore of a lake
e	Epilimnion	5	Water source from groundwater

f	Limnetic Zone	6	Extreme lack of nutrients
g	Mesotrophic	7	Excess nutrients
h	Profundal Zone	8	Region of lake where plants cannot grow from the benthic region
i	Anoxia	9	Sulfur rich, Oxygen Poor
j	Minerotrophic	10	Graph of concentration of species formed from polyprotic acids
k	Oligotrophic	11	Few nutrients
l	Humic	12	Extreme excess of nutrients
m	Ombrotrophic	13	Layer of lake where density changes rapidly
n	Eutrophic	14	Upper water layer in a stratified lake
o	Metalimnion	15	Average amount of nutrients

a. 9
b. 10
c. 12
d. 2
e. 14
f. 4
g. 15
h. 8
i. 1
j. 5
k. 11
l. 6
m. 3
n. 7
o. 13

D. Groundwater

a	Thermokarst	1	opening that leads to underground passages
b	Tower karst	2	pleistocene aged permafrost rich in organics
c	Talik	3	Water source originating from a shaft or cave system under high pressure
d	Ponor	4	Steep-sided flat-bottomed valley terminating in a steep gradient
e	Palsa	5	Concurrent speleogenesis and lithogenesis
f	Blind Valley	6	Domed karstic topography
g	Paleokarst	7	Karstic layer buried by younger rocks
h	Limestone pavement	8	large flat plain surrounding karstic regions
i	Syngenetic Karst	9	Isolated residual hill made of karstic material
j	Mogote	10	mounds of permanently frozen peat and soil core
k	Yedoma	11	Karstic landform with flat surficial limestone with numerous incisions

l	Solution Pipe	12	unfrozen ground in a region with permafrost
m	Kegelkarst	13	Permafrost region with topography heavily shaped by thawing and refreezing of ice
n	Vauclusian spring	14	Karst topography developed beneath a cover of insoluble rock
o	Interstratal karst	15	near vertical drop formed near joint intersections in a karstic environment
p	Pingo	16	Ice cored hills
q	Polje	17	Karst region with large, tall, steep-sided mountains of soluble rock

a. 13
b. 17
c. 12
d. 1
e. 10
f. 4
g. 7
h. 11
i. 5
j. 9
k. 2
l. 15
m. 6
n. 3
o. 14
p. 16
q. 8

E. Human interference

a	Ammonia	1	Low porosity, Low permeability
b	Aquiclude	2	High porosity, high permeability
c	Arch dam	3	NO3-
d	Aquifer	4	Shape of water flow over a ledge
e	Nappe	5	Hollow dam with watertight upstream side supported by material downstream
f	Aquifuge	6	ClO4-
g	Chloramine	7	low barrier across width of river that controls upstream river height
h	Perchlorate	8	NH3
i	Embankment dam	9	Tunnel carrying stream under infrastructure
j	Buttress dam	10	Structure holding up water curved upstream
k	Aquitard	11	R2NCl
l	Culvert	12	Overflow structure designed to prevent overtopping of dam
m	Nitrate	13	(CH3)3OCH3

n	Weir	14	Earthen dam with semipervious waterproof covering
o	Emergency spillway	15	Semipervious geological formation
p	Gravity dam	16	Dam constructed to hold back water using only weight of material used in dam
q	MTBE	17	High porosity, no permeability

a. 8
b. 17
c. 10
d. 2
e. 4
f. 1
g. 11
h. 6
i. 14
j. 5
k. 15
l. 9
m. 3
n. 7
o. 12
p. 16
q. 13

Karst Topography:

1. [multiple select] Karst is a type of topography characterized by the dissolution of:
    a. Limestone
    b. Sandstone
    c. Dolomite
    d. Chalk

Answer: a & c

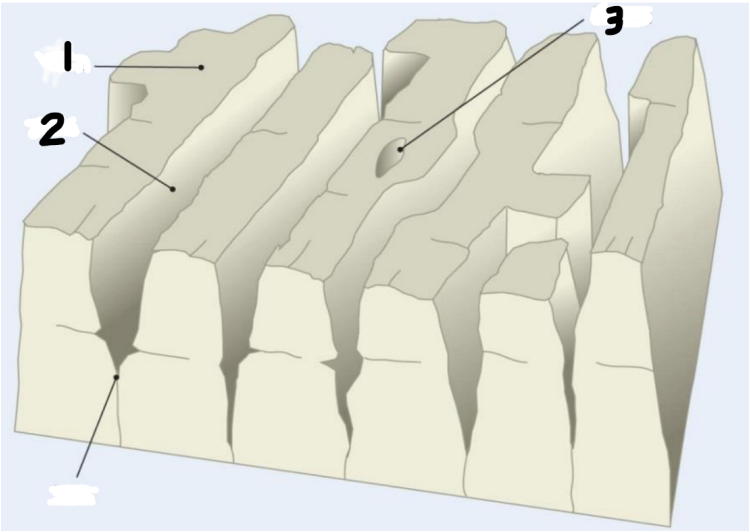
2. [multiple select] Select all of the following conditions that are conducive for karst areas to form:
    a. High water table
    b. Thinly bedded rock
    c. Dense rock
    d. Hardly fractured rock

Answer: b & c

3. [short answer] A tor or castle koppie is a rock outcrop that abruptly rises from gentler hills or ridge crests. What type of rock are tors predominantly made of? Is this an example of a karst formation? In what type of karstic landscape would it be found?

Answer: Tors are predominantly made of granite (also accept: schist, dacite, sandstone). This is not an example of a karst formation (even though it has the form and appearance of one), and is instead labeled as pseudokarst.

4. [multiple select] Limestone pavement is a karst feature of exposed limestone. The following image shows a representation of limestone pavement.



Select all of the following true statements.

a. Feature 1 is an example of a grike, a section of pavement separated from adjacent segments.
b. Grikes develop along joints in limestone.
c. Feature 3 is an example of a karren.
d. Karrens grow in size predominantly through chemical weathering by water.
e. Karrens have the potential to hold a colony of soil in the far future.

Answer: b, d, & e

5. [long answer] The following photo is an example of a helictite, a speleothem that changes its axis from vertical at one or more stages in its growth. What is the most reasonable hypothesis for this behavior? What type of symmetry do helictites have? What specific type of helictites is the photo an example of?



Answer: Helictites are most likely the result of capillary forces acting on tiny water droplets [3], a force strong enough to defy gravity's pull at such a small scale. Helictites possess radial symmetry [2]. The photo is an example of fishtail helictites. [2]

6. [long answer] Regarding the hypothesis in the previous question, in what formation conditions does this hypothesis fully explain helictite formation? What needs to be modified for unfavorable conditions?

Answer: The helictite must have a hollow central tube during formation for water to flow through for capillary forces to have dominance over gravity. [2] However, the majority of helictites are not hollow. For these structures, droplets can be drawn to the tips of many existing structures and deposit their calcite load almost anywhere, leading to winding and irregular patterns. [3]

7.

Standing Water:

1. [Multiple Choice] Which of the following options best explains why spring overturning may occur if warm water tends to be less dense?
    a. Evaporation increases in Spring, which increases salinity and density
    b. Water is densest at around 277K while it remains liquid down to 273K
    c. The lack of an ice layer causes Ekman pumping during the Spring

    d. Melting ice introduces cold, freshwater into the deeper parts of the lake, decreasing its density, allowing easier resurfacing.

Answer: B

2. [Multiple Select] A deep, roughly cone shaped (dimictic) subtropical lake would undergo which of the following processes?
    a. Spring Overturning
    b. Summer Stratification
    c. Fall Overturning
    d. Winter Stratification

Answer: A, B, C, D

3. [Multiple Select] A similar lake as the one in question in the previous question is found in polar regions (~ 75 degrees north) where it is the reason for a surrounding talik. Which of the following processes would the lake undergo?
    a. Spring Overturning
    b. Summer Stratification
    c. Fall Overturning
    d. Winter Stratification

Answer: A, D

4. [Multiple Select] Limnic eruptions do not tend to occur near lakes in direct volcanic contact, but rather some distance away from active volcanic regions. What observation best accounts for this?
    a. Carbon dioxide solubility decreases with temperature
    b. Carbon monoxide solubility increases with temperature
    c. Recurring lake overturning does not allow for gasses to build up in the hypolimnion
    d. Hydrothermal vent fluids cause precipitation of carbonates and deplete the hypolimnion of dissolved carbon dioxide

Answer: A, C

5. [Multiple Select] Which of the following may act as a trigger for a limnic eruption?
    a. Earthquake
    b. Landslide
    c. Meteorological pressure difference inducing a seiche
    d. Rainstorms
    e. Gas development from oversaturation

Answer: All

6. [Multiple Choice] Paludification and terrestrialization refer to opposing processes that lead to the creation of what type of wetland?
    a. Swamp
    b. Marine Mudflat
    c. Estuarine Mudflat
    d. Bog
    e. Marsh

Answer: D

7. [Multiple Choice] A bog with a relatively low pH compared to nearby river fed lakes is likely to be best described by which of the following?
    a. Minerotrophic
    b. Ombrotrophic
    c. Euxinic
    d. Anoxic
    e. Dystrophic

Answer: B

8. [Multiple Select] Select all the following that accurately match the plants that typically dominate the wetland.
    a. Marsh, herbaceous
    b. Swamp, woody plants
    c. Mudflat, herbaceous
    d. Bog, herbaceous
    e. Fen, woody plants

Answer: All

9. [Multiple Select] Vernal pools tend to predominantly undergo which of the following processes annually?
    a. Inundation by excess precipitation
    b. Inundation via flooding from nearby river
    c. Desiccation via drainage by an ephemeral river
    d. Desiccation via seepage into groundwater
    e. Desiccation via evapotranspiration

Answer: A, E

10. [Multiple Choice] What is suspected to be the source of oxygen for large subglacial lakes such as Lake Vostok?
    a. Weathering of bedrock
    b. Decomposition of ancient peat
    c. Air trapped in overlying ice
    d. Influx of surface, oxygen enriched water
    e. There is no source of oxygen; they are largely anoxic

Answer: C

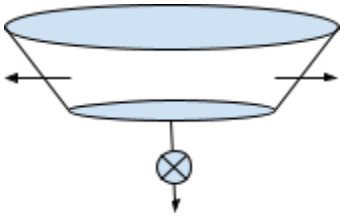
11. [Multiple Choice] Water polluted with large amounts of heavy metals would be precipitated out the slowest in which of the following environments?
    a. Bog
    b. Fen
    c. Mangrove Swamp
    d. Wooded Swamp
    e. Marsh

Answer: A; both can be quite basic

1. [Short Answer] The shoreline development index (SDI) tries to quantize the circularity of a lake by taking the total shoreline length divided by some function of the area. Why can this index only accurately represent lakes of similar size with measurements taken at the same scale?

Answer: coast line problem; they are fractals.

2. An upside down frustum shaped lake with surface area of 6.25 * pi square km, 1 kilometer deep, with a circular bottom at kilometer that is 2 miles in radius. The lake sits on an aquifuge, so no water percolates downwards



After a rainstorm precipitates 3 cm of rain, assuming 10% of the rain becomes runoff, the lake level quickly increases by 10 cm. (Assuming all the water that becomes runoff enters the lake quickly) What is the catchment area of this lake?

Answer: 1/0.1 * 7 / 3 * 6.25 * pi = 458 square km

The lake slowly increases in height in the days after the rainstorm. What would explain this increase in height?

Answer: Interflow water seeping in; flood stage of river

The lake has an outflowing river with a capacity of 3 m^3/s and an inflowing river with a capacity of 10 m^3/s. Assuming the lake is at equilibrium, evapotranspiration to be negligible, K to be 10^-5 (m^2), what is the average hydraulic gradient (assuming the lake to be the top of the water table in the region) around the lateral surface area?

Answer:
Q = AiK
I = Q/(AK) = 7 m^3/s /(1/sqrt(5) * pi * [(5sqrt(5)/2)^2 -(4 sqrt(5)/2)^2] * 1000^2 * 10^-5)
= 0.0443; accept +- 0.04

If the porosity is 0.2, what is the flow velocity of the water leaving the lake as interflow?

q = 7m^3/s / (1/sqrt(5) * pi * [(5sqrt(5)/2)^2 -(4 sqrt(5)/2)^2] * 1000^2 / 0.2

What is the residence time of water in this lake?

A: 15.96976 km^3/10m^3/s = 50.64 years

Flowing Freshwater:

1. [Multiple Select] Which of the following best describes the main difference between Chezy's flow equation compared to Manning's flow equation?
    a. Both are suited primarily for open channel flow
    b. They are suited for open and closed channel flow, respectively
    c. Manning's constant is relatively independent of wetted perimeter
    d. Manning's constant is relatively independent of the hydraulic gradient
    e. Chezy's constant is relatively independent of wetted perimeter
    f. Chezy's constant is relatively independent of the hydraulic gradient

Answer: a, c, d, f

2. [Multiple Select] Which of the following events may lead to river avulsion?
    a. Earthquake leading to significant river aggradation
    b. Flooding events breaching levees
    c. Human activity
    d. Merging of two basins through weathering and erosion
    e. Subsidence or increased deposition in distributaries

Answer: all

3. [Multiple Select] Select all of the following answer choices that are accurate regarding superimposed rivers and antecedent rivers.
    a. Superimposed rivers do not form water gaps
    b. Antecedent rivers do not form water gaps
    c. Superimposed rivers are formed when erosion is faster than uplift
    d. Antecedent rivers are formed when erosion is faster than uplift
    e. Superimposed rivers are formed when erosion of a more resistant barrier is still fast enough to keep up with downcutting
    f. Antecedent rivers are formed when erosion of a more resistant barrier is still fast enough to keep up with downcutting

Answer: d, e

4. [Multiple Select] Select all of the following answer choices that accurately describe a hydrograph.
    a. The rising limb is always steeper than the falling limb
    b. The falling limb is always steeper than the rising limb
    c. The baseflow stays constant
    d. Urbanization usually decreases peak discharge
    e. Urbanization usually decreases lag time
    f. Stream discharge past bank full discharge is largely subsurface movement

Answer: d, e

5. [Multiple Choice] in what section of a river profile would one expect the most rapids and waterfalls?
    a. Headwaters
    b. Floodplain
    c. River mouth
    d. Headwaters and river mouth
    e. Anywhere with metamorphic bedrock

Answer: a

6. [Multiple Choice] Which of the following can be described as a river feature at a knickpoint that creates a large plunge pool that spans the entire width of the river?
    a. Block Waterfall
    b. Kettles
    c. Coulee
    d. Cascade waterfall

Answer: a

7. [Multiple Select] Identify all of the following choices that are accurate concerning the water potential surface.
    a. It is always higher than the water table at any point in an aquifer at equilibrium
    b. It is generally determined by the highest point in the recharge zone of a confined aquifer
    c. Aquifers may not exist above the water potential surface
    d. Cones of depression represent a significant lowering of the water potential surface
    e. Artesian wells are formed when a confined aquifer's water potential surface is higher than the ground surface where the well is dug.
    f. Artificial water potential surfaces for civil water distribution are created by water towers.

Answer: a, b, e, f

8. [Multiple Choice] You are a geologist in the scablands in Washington state when you observe some of the debris seems to have a reversed grading to them - finer sediment is found at higher elevation along a slope and coarse gravels are found at the bottom. Knowing the geologic history of this region (downstream from major glaciers during the last glaciation), best explains these features?
    a. These are examples of Lowe sequences in which an earthquake occurred during a period that stirred the sediment and inverted its deposition sequence.
    b. These are examples of Bouma sequences that have been uplifted.
    c. These are flood deposits in which water rushed up the slope and slowed down which caused inverted bedding.
    d. These were deposited by cascade waterfalls that picked up speed moving downslope and washed away any finer sediment downslope but deposited sediment upslope.
    e. Faulting inverted the relative elevations of the sediment.

Answer: c

9. [Multiple Choice] What type of drainage pattern is associated with eroded domes with strata of alternating resistances, exemplified by the eye of the Sahara?
    a. Radial
    b. Centripetal
    c. Annular
    d. Deranged
    e. Rectangular

Answer: c

10. [Multiple Choice] A paper boat travels at the same speed as a river along the thalweg of a river 4 meters wide. It takes 15 minutes traveling at 0.4 m/s to traverse a straight line distance of 240 meters. What is the approximate sinuosity of the river?
    a. 1.33
    b. 1.50
    c. 1.67
    d. 1.75

Answer: b

11. [Multiple Select] Which of the following are true concerning riverine ice formation and breakup?
    a. Rivers tend to completely freeze over later than lakes
    b. River ice breaks up later at the river mouth than nearer the river source
    c. River ice breaks up later at the river source than nearer the river mouth
    d. The location of first riverine ice break up may depend on direction of flow and latitude.
    e. Ice jams may cause flooding
    f. Ice jams in spring tend to be larger in scale than freeze-ups in the beginning/middle of winter

Answer: a, d, e, f

12. [Multiple Choice] What is the main source of the **naturally** occurring dissolved load in sediment?
    a. Ions from groundwater
    b. Ions from rainfall
    c. Ions from runoff
    d. Ions from biological weathering of rocks on the river bed
    e. Chemical leaching from rocks found on the river bed

Answer: a

1. [Short Answer] Referencing the Hjulstrom curve, around what value on the Krumbein Phi scale is sediment that are the easiest to erode?

Answer: accept range 2 → 4

2. [Short Answer] Expressed in an arbitrary radius r and assuming it is completely filled, what is the hydraulic radius of a circular pipe? Show some work if possible.

Answer: pi*r^2 / (2*pi*r) = r/2

3. [Short Answer] 3 part question: look out for limits for water height (genuinely there to make it easier for you :DD)

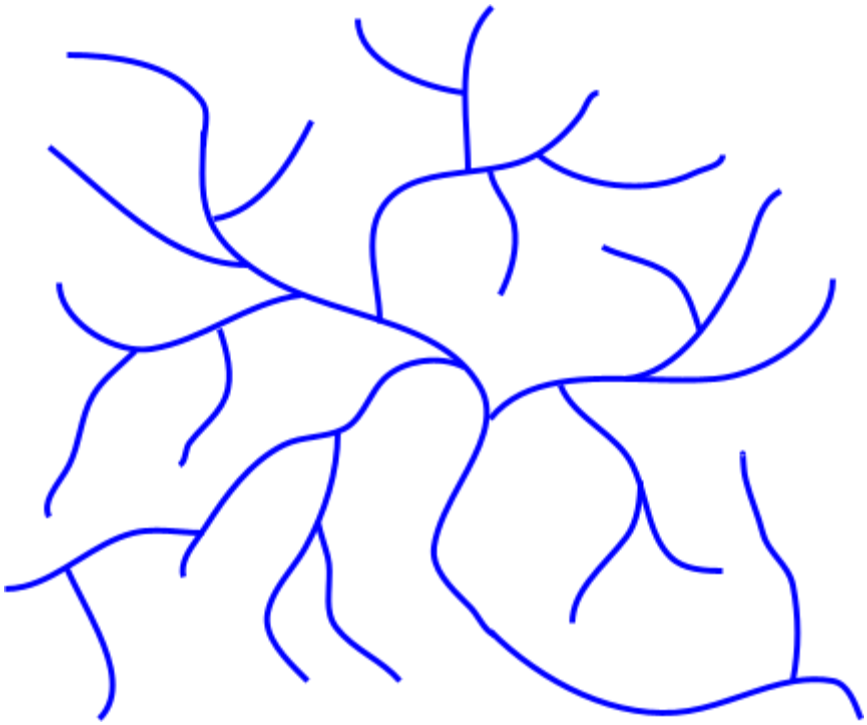
   Expressed in an arbitrary length L and assuming it is filled to some height H, what is the hydraulic radius for a square pipe laid on one of its flat sides (assume H < L)?

   For the same square pipe flipped 45 degrees onto one of its vertices (assume H < L/sqrt(2))?

   For a semicircular pipe with arbitrary radius R and water height H (Assume H << R; please use small angle approximation — sin(x) = x)? Do show work.

Answer: (H*L)/(2H+L); sqrt(2)/4 * H; approximates H/2

4. [Short Answer] {Stream segments meaning sections that may have a different stream order depending on the system} The hypothetical system drains to the bottom right.



Using Strahler ordering: [Feel free to label on the diagram; that won't be graded]
How many second order stream segments are present in the above diagram?
How many third order streams are present in the above diagram?
What is the order of the trunk stream in the above diagram?

Using Shreve ordering:
How many first order stream segments are present?
How many second order stream segments are present?
What is the order of the trunk stream in the above diagram?

Using Hack ordering:
What is the order of the trunk stream in the above diagram?
How many of the highest order stream segments are present?

What type of drainage pattern does this hypothetical river basin roughly approximate?

Answer: 13, 4, 4; 22, 8, 22; 1, 3; dendritic

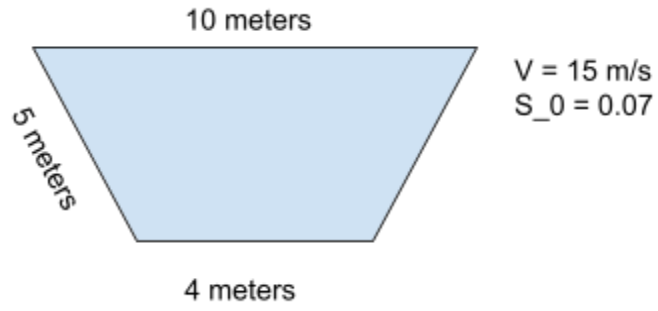
5. [Short Answer] What is the maximum flow rate of an open semicircular pipe (no top) with a radius of two meters, Manning roughness coefficient of 0.015

s/m^(⅓), acting as an emergency water release channel for a small dam that funnels water down 5 meters of relief over 100 meters of distance?

Answer: 93.7m^3/s

(2pi/2pi)^⅔ * 0.05^0.5/0.015 * 2pi
Last 2pi bc flow rate not flow velocity.

6. [Short Answer] What is the Manning roughness coefficient of a top-open canal with the following cross section? Assume the canal is full.

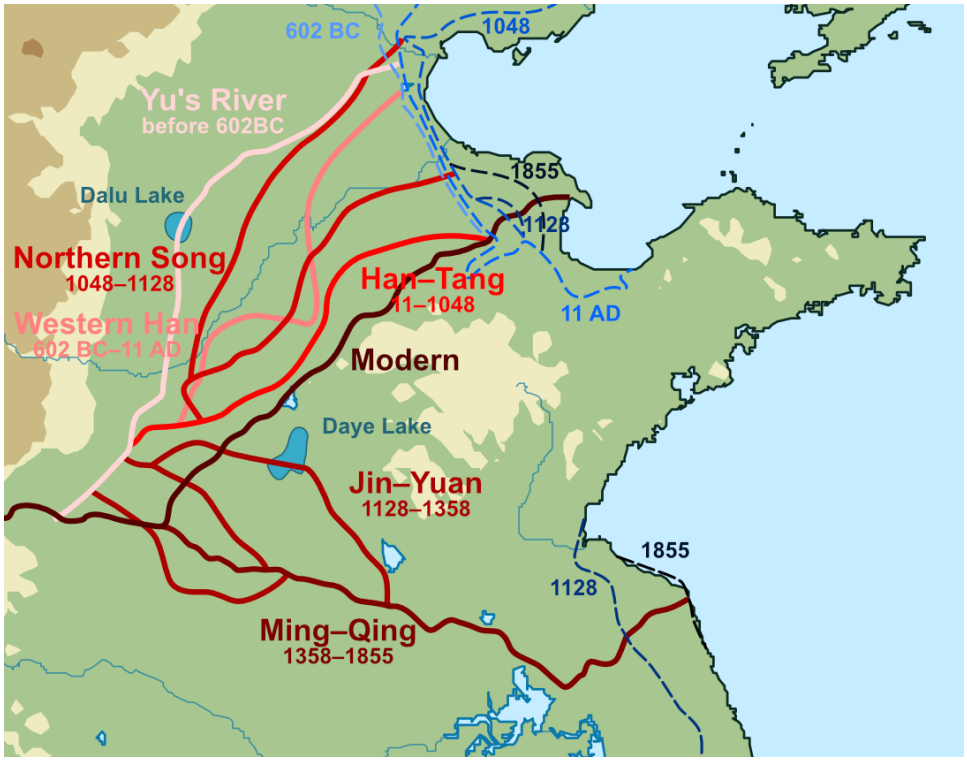


Answer: 0.028. No units required, but if units are given, must be sec/m^⅓
sqrt (0.07) *(28/14)^(2/3) /15
14 bc top open.

7. [Short Answer] identify the type of stream by the description given.
    a. Only flows after plenty of rainfall
    b. Only flows seasonally
    c. Fully above the water table
    d. Fed by water from the saturated zone
    e. Completely underground

Answer: ephemeral, Intermittent, losing, gaining, subterranean

1. [Long Answer] In the past 3000 or so years, multiple causes have resulted in the dramatic changes in the course of the lower reaches of the Yellow River (and even caused changes in the flow of other rivers). Explain how this may happen in at least two different (if not slightly related) ways, with at least one way human impact may influence the changes in the river.

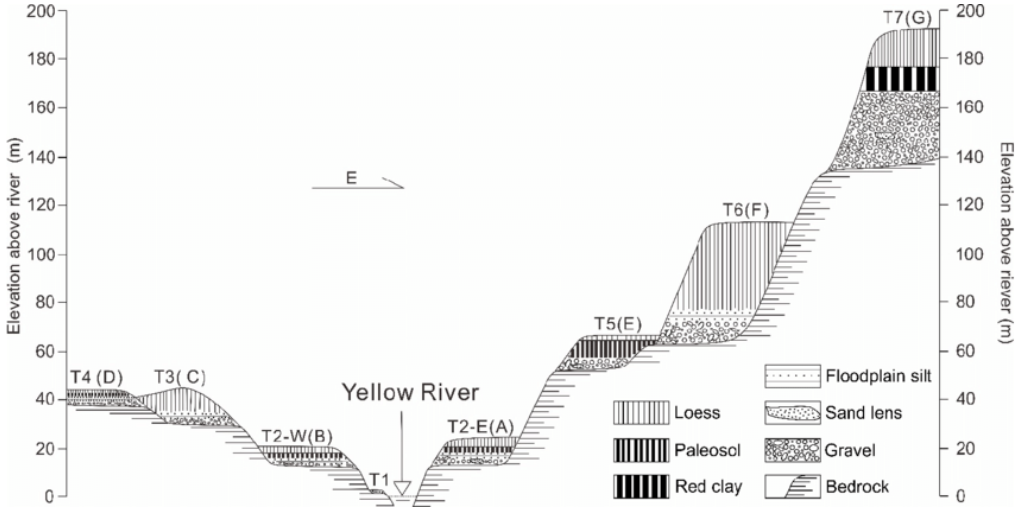


Uplift of Tibetan Plateau increases the relief water travels down. Earthquakes may introduce instability into levees and general infiltration during higher stages leads to the possibility of levees collapsing, leading to outbreak and flooding. General deposition of sediment within the river channel also increases the flood stages over time and leads to overtopping of the levees eventually. [2 pts each for levee collapse and channel fill]
Human impact by constructing dams and increasing levee heights may exacerbate the severity of flooding due to promoting sediment deposition.
[2 pt for human impact]

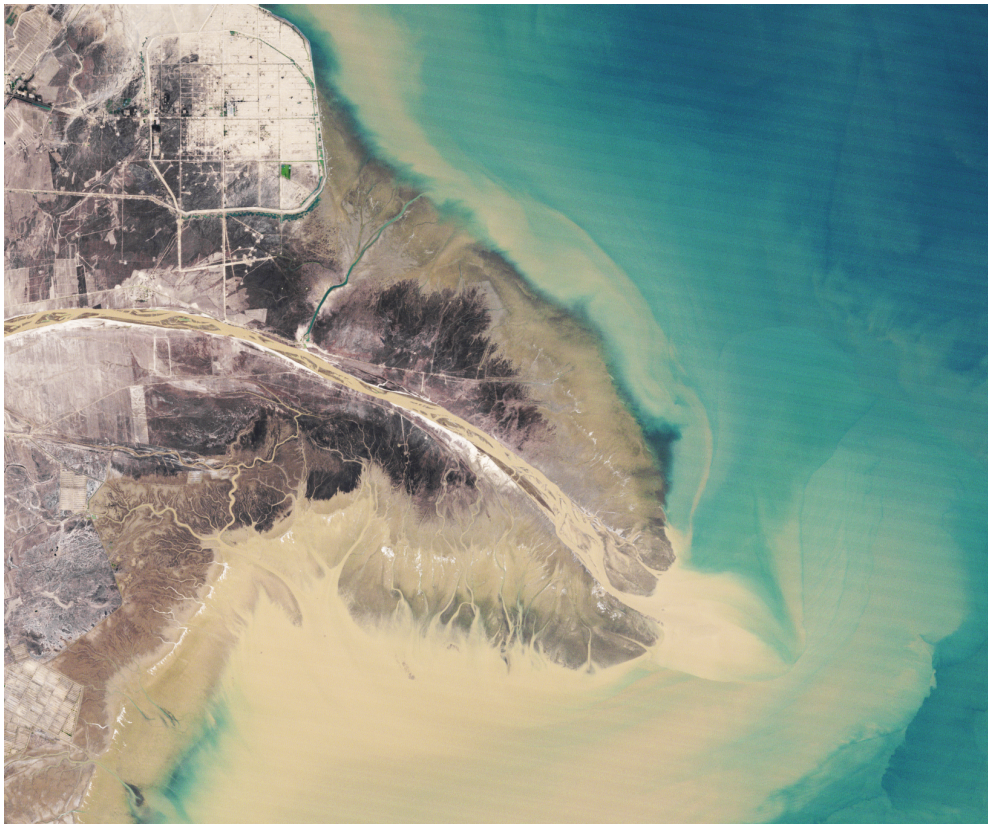
2. [Label] Identify which of the following terraces may be identified as:
   Strath Terraces:
   Fill Terraces:
   Paired Terraces (write them in pairs with parentheses around them):
   Active floodplain:

A: All but T7 and T1 are strath terraces; T7 is a fill terrace. (T2-W, T2-E)
AF: T1
[1 pt for each thing correct; not marking T2 as both paired and strath is -1]

Identify the indices of the key as either fluvial/aeolian or neither in origin.
Loess: Aeolian
Paleosol: Neither/Aeolian; both ok
Red Clay: Neither/Aeolian; both ok
Floodplain Silt: Fluvial
Sand lens: Fluvial
Gravel: Fluvial
[1 pt each]



3. [Short Answer] What type of delta is this best categorized as?



A: River dominated Delta [2]

4. [Multiple Choice] What event has just occurred here? This photo was taken around 5 years after the previous question. What would be expected to cause this event to occur?



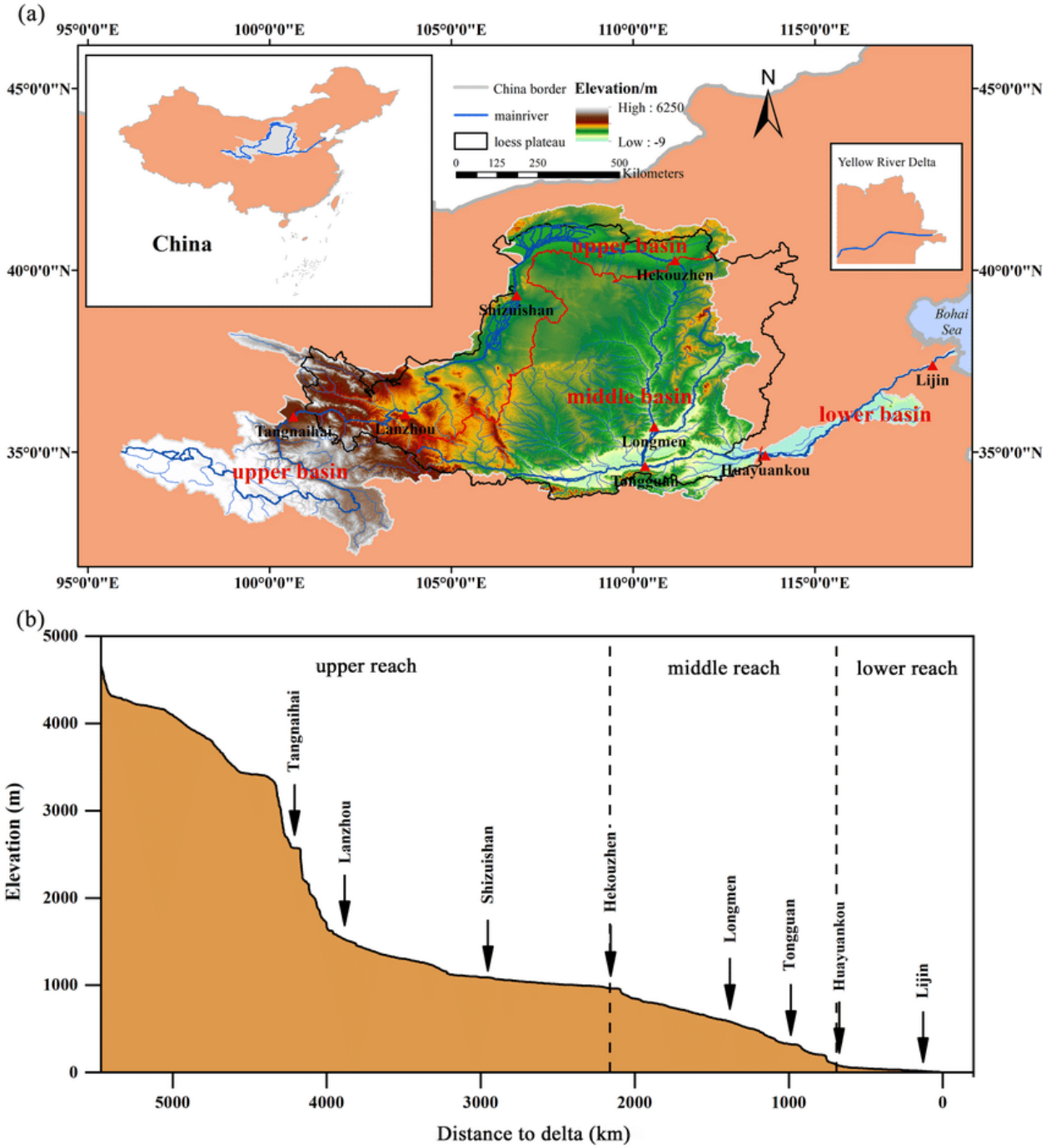
A: Avulsion [2] ; disturbance to distributary caused breaking in levees and forming another main distributary [2]

5. [Long Answer] Why is flooding such a problem along the Yellow River? Even though it doesn't flow to the sea perennially due to heavy agricultural usage, it still may pose a flood risk. Give at least 3 reasons and explain each thoroughly.

   B. Give two methods of flood control that can be implemented and explain their shortcomings in this particular case due to the river's high sediment load.

   Answer: High sediment load → overfill levees more easily [3], risk of earthquakes + uplift leading to periodic flooding [3], large seasonal variability in water output due to ice melt. [3]

   Dams and levees [1 each]; filled up due to sediment [2]


(a)
95°0'0"E
100°0'0"E
105°0'0"E
110°0'0"E
115°0'0"E
45°0'0"N
40°0'0"N
35°0'0"N
China
China border
mainriver
loess plateau
Elevation/m
High : 6250
Low : -9
0 125 250 500
Kilometers
N
Yellow River Delta
upper basin
Hekouzhen
Shizuishan
Bohai Sea
Lijin
middle basin
lower basin
Tangnaihai
Lanzhou
Longmen
Tongguan
Huayuankou
upper basin
(b)
upper reach
middle reach
lower reach
Tangnaihai
Lanzhou
Shizuishan
Hekouzhen
Longmen
Tongguan
Huayuankou
Lijin
Elevation (m)
5000
4000
3000
2000
1000
0
Distance to delta (km)

Human interaction and Critical Zone

1. How many meters of freshwater is present beneath a coastal zone where saltwater intrusion is present and the water table is 2 meters above sea level? Assume seawater density to be 1.03g/cm^3 and freshwater to be 1.00g/cm^3.

Answer: 2 + 2/0.03 = 68.7777 → 69m [5]

2. An insurance agent poorly understands probabilistic flooding. By their calculations, they believe a house they are insuring for $50,000 at a rate of $1,000 per year is within a 100-year flood zone is the optimal pricing since after 50 years they believe there is a 50% chance for the insurance company to need to pay out. However, you know this is not necessarily the case. After how many years would this house have a 50% chance to flood?

Answer: N = ln(1-0.5)/ln(0.99) = 69 years [5]

3. Flash flooding is much more of a concern in regions that went through prolonged drought. Which of the following best explains the most important factor?
    a. Decrease in vegetation associated with drought leads to decreased water absorption.
    b. During a drought, bacteria create specific proteins and lipids that are highly hydrophobic and decrease infiltration rate
    c. Droughts lead to compaction and decrease in topsoil porosity and permeability
    d. Dried ground becomes more hydrophobic since its surface area decreases
    e. Droughts lead to more aeolian erosion of clays leading to decreased permeability

Answer: C [2]

4. Darcy's Law would be valid for which of the following situations?
    a. Water flowing through an aquifer
    b. Honey flowing through a series of thin horizontal pipes
    c. Airflow over an aircraft wing in flight
    d. Water flow around the bow of a ship

Answer: A [2]

5. [multiple select] Which of the following correctly compares open channel and pipe flow?

a. Open channels flow because of gravity while pipes flow under hydraulic pressure.
b. Both open channels and pipes can be of any shape.
c. The piezometric head of an open channel varies with the pressure of the flow.
d. The piezometric head of pipe flow varies with the length of the pipe.
e. The location of maximum velocity of an open channel and pipe flow are both dependent on the channel's roughness.

Answer: a [2.5]

6. [multiple choice] The hardest part of using Chezy's equation is selecting the correct value for the Chezy constant C. The Bazin Formula provides an empirical relation for C dependent on R (the hydraulic radius) and K (the Bazin constant). Which of the following materials would have the highest K value?
    a. Masonry rubble
    b. Concrete
    c. Dredged earth channels
    d. Brickwork

Answer: c [2]

1. Explain how dams change the downstream environment. Be sure to include any forms of pollution and the changes that the impoundment of water may cause.

Answer: thermal pollution due to release of primarily surface or deeper water → both lead to thermal changes; sedimentation behind dam and increased erosion downstream of dam; increased evaporation from increased surface area. (points given for other correct answers relating to other aspects such as fish/wildlife, etc).

2. [Short Answer] At an elevation of 400 meters, the water table is 30 meters below the ground. At an elevation of 350 meters, the water table is 20 meters below the ground. The straight overland distance between the two points of measurement is 2 km. What is the average hydraulic gradient assuming the water table in this region is in some sort of equilibrium?

    A: (400-30-(350-20))/sqrt(2000^2-50^2) = 0.02